

An Introduction To Reverse Mortgages

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- ▶ Reverse mortgages are loans given to senior citizens against the value of their homes. The reverse mortgage is primarily designed for "house rich, cash poor" elderly homeowners. The entire mortgage loan is due upon occurrence of certain maturity events such as death of the borrower (mortality) or the borrower's moving out of their home for a period of more than twelve months (mobility). Reverse mortgage borrowers do not have to make regular monthly payments.
- ▶ Due to the projected rise in living and healthcare costs, uncertainty surrounding Social Security benefits, and the precipitous decline in the number of workers belonging to a defined benefit pension plan, it seems likely that senior citizens will find it increasingly difficult to maintain their living standards based on their savings alone. Housing is the largest asset for most senior citizens and under the scenario outlined above, they are expected to increasingly tap into their home equity to maintain living standards. This should be a driving factor for the increased popularity of reverse mortgages in the near term.
- ▶ The most important type of reverse mortgage in the U.S. is the Home Equity Conversion Mortgage (HECM). The primary focus of this paper is to present a detailed description of HUD's Home Equity Conversion Mortgage (HECM) program. The HECM program has a greater than 90% share of the total U.S. reverse mortgage market and is the only program in which the reverse mortgage is insured by the federal government. We discuss some of the attributes of borrowers who take out loans under the HECM program along with an analysis of their loan characteristics. This is followed by an analysis of the repayment and draw behavior of these loans. We conclude with a brief account of some of the relevant factors to consider in a typical HECM securitization.

I. Introduction

Reverse mortgages allow homeowners to convert home equity into cash without selling their houses or having to make monthly payments.

A reverse mortgage is a loan advance to senior citizens against the value of their homes. The reverse mortgage allows the homeowner to convert their home equity into cash without selling the house or making monthly payments. The reverse mortgage is primarily designed for "house rich, cash poor" elderly homeowners. A borrower has the option of choosing one of the following payment plans: take a lump sum payment, establish a line of credit to tap into, receive fixed payments from the lender over the course of their life, or receive fixed payments over a pre-selected term. A borrower does not need to repay the loan until they die (in which case the borrower's estate pays the loan), move out of their home for a period of twelve months, or sell the house.

Demographic trends will likely contribute to increasing demand for reverse mortgages.

We believe that there are several factors that will drive increasing demand for this product going forward. First, the population base towards which this product is targeted is projected to grow very rapidly over the next several years. With the baby boomers beginning to enter retirement, the age group of 65 years or older is projected to double over the next 20+ years (see Figure 1). The U.S. Census Bureau estimates a homeownership rate of 80% for households in which the age of the homeowner is 65 years or older. Due to the projected rise in living and healthcare costs, uncertainty surrounding Social Security benefits, and the precipitous decline in the number of workers belonging to a defined benefit pension plan, it seems likely that a fair number of senior citizens will find it increasingly difficult to maintain their living standards based on their savings alone. Also, it seems reasonable to assume that the baby boomers will increasingly tap into their home equity to maintain their present living standards. Reverse mortgages will appeal to such house rich, cash poor seniors over other competitive products (such as a home equity line of credit, HELOC for short) because the borrower does not have to make any monthly payments. In addition, a reverse mortgage also offers the convenience of allowing all interest and fees to be rolled up into the loan balance.

The FHA-backed Home Equity Conversion Mortgage (HECM) program is the largest reverse mortgage program in the U.S.

Many institutions, including the U.S. federal government, some local and state governments, and banks and mortgage companies are currently involved in issuing reverse mortgages. One of the largest programs created for the purpose of facilitating the origination of reverse mortgages is the Home Equity Conversion Mortgage (HECM) program backed by the Federal Housing Administration (FHA), which is part of the Department of Housing and Urban Development (HUD). The HECM program has a greater than 90% share of the total U.S. reverse mortgage market and is the only program in which the reverse mortgage is insured by the federal government. The program provides a guarantee to borrowers that lenders will meet their obligations¹ while simultaneously insuring lenders against borrower default. The HECM program is available in all 50 states plus the District of Columbia and Puerto Rico. Figure 2 shows the number of HUD-guaranteed reverse mortgages originated since 1990. The increase in originations over the past several years is an indication of the growing demand for this product in the market place. The other types of reverse mortgage programs in the U.S. are Fannie Mae's *Home Keeper* program and those established by private lenders. This primer

¹ Besides servicing the loan, lenders are also responsible for sending cash advances (either monthly or lump sum payments) to reverse mortgage borrowers.

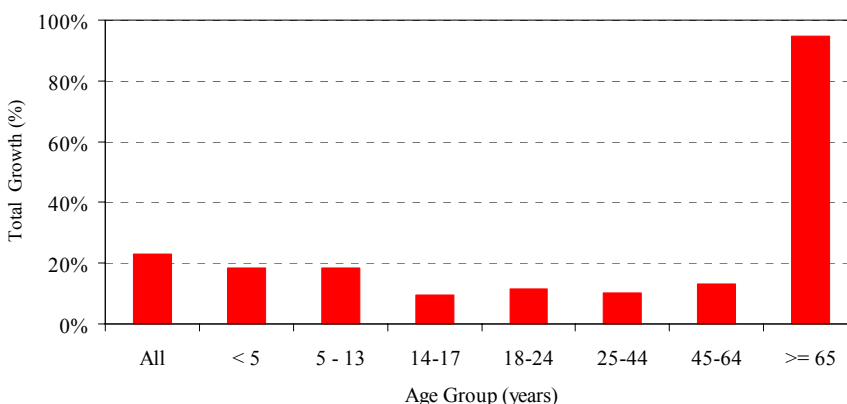
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focuses on the HECM program because of its overwhelming share of the U.S. reverse mortgage market.

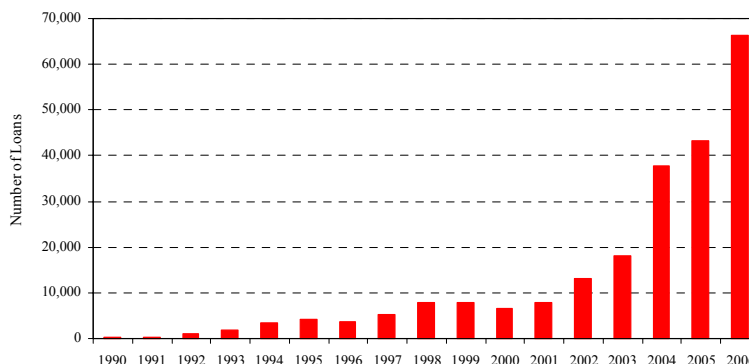
The organization of this paper is as follows. Section II describes various features of the HECM program. Section III provides a detailed breakdown of the characteristics of borrowers and loans originated under the HECM program. Section IV discusses the repayment and draw behavior of HECM loans. A few issues that are relevant to securitization of HECM loans are discussed in Section V.

Figure 1: U.S. Population Growth Estimates by Age Group: 2005 - 2030



Source: U.S. Census Bureau

Figure 2: Annual Origination Volume for HECMs by Fiscal Year*



Source: HUD.

*The Federal Fiscal Year (FY) begins on October 1st and runs through September 30th of the following year.

II. The Home Equity Conversion Mortgage (HECM) Program

The HECM program was established in 1987 by Congress and formally kicked off in July 1989. At first, the program was run on a pilot basis to see if a mortgage that allowed the conversion of home equity into cash would be popular among elderly homeowners. The success of the program led to it being established on a permanent basis in 1998 albeit with a cap on the number of outstanding reverse mortgages insured by the program. The cap is currently at 250,000.

HECM loans are available to borrowers over the age of 62.

A HECM loan is designed for borrowers over the age of 62 and can either be taken alone or with a spouse as the co-borrower. There is no monthly payment requirement on the loan and the loan is due in full only on the death of the last of the two co-borrowers or if the borrowers do not occupy the home as their primary residence for a period of greater than 12 months. Borrowers have the option to repay the loan at any time either in part or full without a penalty. However, the high upfront costs associated with refinancing a HECM loan discourage prepayments. After the death of a borrower, their heirs do not need to sell the property to pay back the loan and can repay the loan from their other assets. We now go through a detailed description of the various features of the HECM program.

What are the eligibility requirements for a HECM?

1. The age of each borrower has to be at 62 years or older and the reverse mortgage needs to be backed by their principal residence. There are no restrictions on borrower income.
2. Generally speaking, the borrower(s) cannot have any currently outstanding debt on their home. If there are any such debts, they will need to be paid off either before the homeowner receives the reverse mortgage or with the money from the reverse mortgage. Consequently, a reverse mortgage is considered a "first lien" against the home.
3. In addition to age and ownership requirements, each borrower must not be delinquent on any federal debt or, with some exceptions, should not have had a claim paid on a HUD-insured property/loan within the past three years.
4. The home must either be a single-family residence in a 1- to 4- unit dwelling, part of a planned unit development (PUD), or a HUD-approved condominium. Cooperatives and most mobile homes are not eligible but certain manufactured housing could qualify.
5. The home must meet HUD's minimum standards and be at least one year old. Borrowers can use proceeds from the HECM loan to pay for any required home repairs.
6. HECM borrowers are required to discuss the program with a counselor from a HUD-approved counseling agency before filing an application for the loan.

There is no income-based eligibility requirement for reverse mortgages.

When does a borrower repay the loan?

There are no required repayments on a reverse mortgage unless one of the following *maturity events* occurs:

No repayment on a HECM loan is required unless a maturity event occurs.

1. The last surviving borrower dies (mortality);
2. The borrower sells their home (mobility);
3. The borrower transfers the title of the home to someone else; or
4. The borrower has not lived in the home for one continuous year.

Note that the borrower is responsible for property taxes, insurance, and general maintenance of the property. A reverse mortgage lender can require repayment if these conditions are not satisfied. Typically, lenders have the option to pay for these expenses and roll them into outstanding balance of the mortgage.

The entire principal on a reverse mortgage is due when a maturity event occurs.

When a *maturity event* occurs, the lender must be repaid the entire outstanding loan balance in one single payment. The borrower or their heirs (or estate) can pay the loan proceeds either by selling the house or from any other source of cash. For instance, the borrower or their heirs can take out another regular mortgage on the home to pay off the reverse mortgage loan. Note that a borrower can choose to make partial repayments on a reverse mortgage loan without any penalty. A full repayment by the borrower will terminate the loan agreement. Also note that when a maturity event occurs, it is possible for the outstanding loan balance to exceed the *maximum claim amount* (defined later in this section) either because of the longevity of the borrower or a higher than expected interest rate. However, the borrower or their estate is not liable for any shortfall not covered by the value of the home.

How much can a reverse mortgagor borrow?

FHA guarantees that HECM lenders will be repaid the entire outstanding balance at the occurrence of a maturity event.

Since a borrower does not have to make any regular monthly payments on their loan, the outstanding balance on a reverse mortgage will grow over time but the total debt is considered *non-recourse*. Conceptually, this means that the lender has only the borrower's home to make up for any shortfall in the repayment of the outstanding balance whenever a maturity event occurs. For HECM loans, FHA provides a guarantee that lenders will be reimbursed for any repayment shortfall when a maturity event occurs. Therefore, there is a *maximum claim amount* attached to the non-recourse debt that is lesser of the appraised value of the borrower's home and the maximum loan amount that can be insured by the FHA for residences in a given geographic area. For instance, if the appraised value of a borrower's home is \$600,000 but the borrower lives in an area where FHA will insure loans only up to \$200,000 then the maximum claim amount will be \$200,000. Figure 3 shows the current FHA mortgage limits for different types of homes. In 2006, for a one-family home, the FHA loan limit varied from \$200,160 for basic standard mortgages to \$362,790 for high-cost areas.

Figure 3: FHA Mortgage Limits

	One-Family	Two-Family	Three-Family	Four-Family
Standard Mortgage	200,160	256,248	309,744	384,936
High Cost Areas	362,790	464,449	561,411	697,696
Alaska, Guam, Hawaii & Virgin Islands	544,185	696,673	842,116	1,046,544

Source: HUD

The *principal limit* is the maximum cash available to borrowers at origination.

The maximum claim amount is used to determine the amount of cash (*principal limit*) a borrower can receive at origination. The calculation of the principal limit of a reverse mortgage depends on the age of the youngest borrower, the maximum claim amount, and the *expected average mortgage interest rate*. Figure 4 illustrates the cash available to a reverse mortgage borrower assuming a maximum claim amount of \$100K at different assumed

interest rates. All else being equal, the older the borrower is, the larger their principal limit. The reason for this is that the life expectancy of a borrower decreases with age, which increases the chances of a lender recovering their principal in a shorter time.

Figure 4: Principal Limit Table For a Maximum Claim Amount of \$100,000

Borrower Age	Average Mortgage Interest Rate		
	7%	8%	9%
62	\$45,700	\$37,100	\$30,200
65	\$48,900	\$40,500	\$33,600
70	\$54,700	\$46,700	\$40,000
75	\$60,900	\$53,700	\$47,200
80	\$67,400	\$61,100	\$55,300
85	\$73,800	\$68,600	\$63,600
90	\$79,800	\$75,700	\$71,700
95	\$85,900	\$83,200	\$80,500

Source: Banc of America Securities

Note that if the rate at which the mortgage accrues interest is greater than the rate of home price appreciation (HPA), then a crossover point exists beyond which the value of the outstanding loan is greater than the value of the (appreciated) home. Since younger borrowers are expected to live longer, there is a greater likelihood that their outstanding loan balance could grow larger than the value of their (appreciated) home. Consequently, the principal limit is usually smaller for younger borrowers and higher for older borrowers. HECM lenders are able to assign a reverse mortgage to FHA when the outstanding balance is at least 98% of the maximum claim amount. Thus, the risk that the outstanding balance of a reverse mortgage loan will be greater than the value of the home is borne by FHA rather than by HECM lenders.

Note that Figure 4 employed a fixed long term rate (*expected average interest rate*) to determine the borrower's principal limit at closing. The lower the expected average interest rate, the lower the expected cost of borrowing, and the more funds available to the borrower. The expected average interest rate used to calculate the HECM principal limit is based on the sum of the 10-year CMT rate, a margin and a 50 bps monthly insurance premium, rounded to the nearest 1/8%. This rate has nothing to do with the actual interest rate (*applied interest rate*) charged to the borrower. The HECM rate is indexed off of 1-year CMT and can adjust either monthly or annually.

Unlike some of the other reverse mortgage programs (such as Fannie Mae's Home Keeper) in which the principal limit remains constant over time, the principal limit available to borrowers (at origination) in the HECM program is increased monthly by the applied mortgage interest rate. Consequently, any credit line remaining after the loan closes will grow over time, a feature not available in most other reverse mortgage programs. Also, the monthly adjustable HECMs come with a lower margin which means that the principal limit available to a borrower under this plan is larger than the one available under the annually adjustable plan. This is partly responsible for the overwhelming popularity of monthly adjustable HECM loans over annually adjustable ones.

The *principal limit* is increased monthly by the applied mortgage interest rate. Consequently, any credit-line remaining after the loan closes will grow over time.

For a monthly adjustable reverse mortgage, borrowers are typically charged an interest equal to 1-year CMT + 150 bps margin + 50 bps MIP.

Borrowers pay a mortgage insurance premium (MIP) that consists of an upfront fee of 2% on the maximum claim amount plus a 50 bps annual fee (charged monthly) on the outstanding loan balance.

What are the interest rate and fees charged on the loan?

As previously noted, the interest rate charged on an HECM loan is adjustable and is based on a spread (margin) over the weekly average 1-year CMT rate with a lookback period of 30 days. HECM borrowers can choose either an annually adjustable mortgage rate having a reset limit of 2% in any year and a life cap of 5% over the initial rate, or a monthly adjustable mortgage rate without any monthly reset limit and a life cap of 10% over the initial rate. At present, the margin on HECM loans is 1.5% for loans with a monthly adjustment and 3.1% for ones with an annual adjustment. About 88% of the HECM borrowers select the monthly adjustable rate option. The borrower's choice between a monthly adjustable or an annually adjustable loan is irrevocable and cannot be changed unless they refinance.

The total origination fees for a reverse mortgage can vary across lenders and may include fees for property appraisal and inspection, standard closing costs, origination fees, points, mortgage insurance premiums, and monthly servicing fees. Lenders typically allow these costs to be financed by rolling them into the outstanding loan balance. For HECM reverse mortgage loans, the origination fee is capped at the greater of \$2,000 or 2% of the *maximum claim amount*.

The FHA guarantee on HECM loans is financed by the *Mortgage Insurance Premium (MIP)* which consists of a non-refundable 2% upfront premium charged on the maximum claim amount and a 50bp annual fee (charged monthly) on the outstanding loan balance. The borrower has the option to finance the entire upfront mortgage insurance premium.

Borrowers of HECM loans are also charged a flat monthly servicing fee on their loan. The FHA monthly limit on the servicing fee is \$30 for annually adjustable interest rate loans and \$35 for monthly adjustable loans. The lender typically finances this monthly servicing fee by setting aside a portion of the *principal limit* at origination. This set aside amount can total several thousand dollars but is just a calculation and not an actual fee. The servicing fee is charged to the borrower by increasing the outstanding balance of the loan on a monthly basis.

In general, when all these fees are added up, HECM origination fees currently range from 2.5% to 5% of the loan amount. This is quite high compared to the situation for regular "forward" mortgages where loan fees typically average 1% of the loan amount.

How are the loan proceeds paid to the borrower (payment plans)?

The payment received from a reverse mortgage can either be a single lump sum, a regular monthly cash advance (for a fixed term, or for as long as the borrower is alive and maintains their primary residence at home), a line of credit, or some combination of these payment methods. Specifically, the HECM program offers borrowers the following five payment plans:

Most HECM borrowers select a line of credit as their payment plan.

1. *Tenure*: A borrower receives equal monthly payments until a maturity event occurs.
2. *Term*: A borrower receives equal monthly payments only for a specified time period. However, note that the borrower does not have to repay the loan until a maturity event occurs. Monthly payments under the term plan are higher than the payments under the tenure plan and increase with a decrease in the term selected or with the borrower's age. Figure 5 shows monthly cash advances available to borrowers of different ages assuming a maximum claim amount of \$100K and a 7% expected interest rate.
3. *Line of Credit*: This allows a borrower to establish a line of credit equal to their principal limit less certain loan origination, closing, servicing and MIP fees (assuming that the borrower rolls these costs into their outstanding balance). A borrower can draw from their line of credit at will as long as they do not exhaust their credit line and any maturity event requiring repayment does not occur. The unused portion of the borrower's line of credit grows over time and the borrower is only charged interest against the amount drawn by them. The line of credit (LOC) is the most popular payment plan for HECM borrowers.
4. *Modified Term*: This is a term plan combined with a line of credit. The borrower can elect to set aside a portion of their principal limit to establish a line of credit and receive the remaining principal over a selected term in equal monthly payments.
5. *Modified Tenure*: This is a tenure plan combined with a line of credit. The borrower elects to set aside a portion of their principal limit to establish a line of credit and receives the remaining principal in equal monthly payments until a maturity event occurs.

Figure 5: Monthly Cash Advance (Maximum Loan Amount of \$100K & 7% Interest)

Borrower Age	Term In Years		
	Tenure	10 Year	5 Year
62	\$285	\$528	\$900
65	\$311	\$564	\$963
70	\$362	\$631	\$1,077
75	\$428	\$702	\$1,199
80	\$520	\$778	\$1,327
85	\$659	\$852	\$1,453

Source: FHA, Banc of America Securities

Having described some of the salient features of HECM loans, we now present an illustrative example to further reinforce the concepts described earlier (see Figure 6). We assume that the borrower is a 65 year old male, lives in San Francisco, and has a house with an appraised value of \$500,000. Since the maximum amount that FHA will insure for a single-family home in San Francisco is \$362,790, the borrower's *maximum claim amount* is also \$362,790. The current loan interest rate will be 7.17% (1-year CMT + 150 bps margin + 50 bps monthly MIP) if the borrower chooses the monthly adjustable loan option and 8.77% if the borrower elects for the annually adjustable option. Likewise, the borrower's *principal limit* at closing,

which is the maximum amount of credit available to the borrower at loan closing, will depend on whether the borrower chooses the monthly or annually adjustable loan option. His principal limit will be \$198,809 for a monthly adjustable loan and \$147,656 for the annually adjustable loan. Note that the borrower's principal limit will grow over time with a current annualized growth rate of 7.41% (9.13%) for the monthly (annually) adjustable loan. In the example shown in Figure 6, we assume that the borrower elects to roll all the loan origination and insurance costs into his outstanding balance. This reduces the amount of cash or credit line available to the borrower. Let's assume that the borrower establishes a line of credit for \$100,000 and elects to receive monthly cash payments for tenure on the remaining cash balance. His monthly cash payments will be \$497 under the monthly adjustable loan and \$203 under the annually adjustable option.

Note that once a borrower decides between a monthly adjustable and an annually adjustable rate option, he cannot switch his option unless he refinances his loan. However, HECM lenders provide borrowers complete flexibility in changing their payment plan and a borrower can switch from one payment plan to another by paying a small administrative fee of no more than \$20. The borrower does not have to pay the loan origination and closing costs again if he decides to switch his payment plan.

In the second half of Figure 6, we show the various payment plan choices available to the borrower. For the monthly adjustable rate option, if the borrower elects to take a single lump-sum advance of \$177,297, then he does not have any remaining credit-line, i.e., the borrower has withdrawn 100% of his available credit-line at inception. In such a scenario, the borrower's outstanding balance at loan origination is \$194,143 (principal limit of \$198,809 less servicing fees set aside of \$4,666). This assumes that at origination the borrower does not pay in cash the loan origination fees, closing costs, and mortgage insurance premiums but instead rolls these fees into his outstanding balance. Also, note that the servicing fees set aside are not counted against the borrower's outstanding loan balance at inception but accrue each month as the fees are earned.

Instead of taking a lump-sum cash payment, the borrower can establish a credit line of \$177,297 that grows each year at an annualized rate of 7.41%. The borrower can also elect a tenure payment option under which he receives a monthly cash payment of \$1,140 until a maturity event occurs. Finally, the borrower has the option to select any combination of lump sum payment at closing, credit-line account, and monthly cash payments.

Figure 6: Illustrative Example for a 65 Year Old HECM Borrower

Loan Calculations	Monthly Adjusting Loans	Annual Adjusting Loans
Current interest rate index (1)	5.17%	5.17%
Lender's margin	1.50%	3.10%
HUD Mortgage Insurance	0.50%	0.50%
Current loan interest rate	7.17%	8.77%
Annual growth rate in creditline	7.41%	9.13%
Cap on effective loan rate	17.17%	13.77%
Value of the borrower's home	\$500,000	\$500,000
FHA Lending Limit	\$362,790	\$362,790
Maximum Claim Amount (2)	\$362,790	\$362,790
Loan Principal Limit	\$198,809	\$147,656
Less loan fees to lender	\$7,256	\$7,256
Less mortgage insurance (3)	\$7,256	\$7,256
Less other closing costs	\$2,334	\$2,334
Less service fees set aside	\$4,666	\$4,666
Cash available to the borrower	\$177,297	\$126,144
Less desired creditline	\$100,000	\$100,000
Left for monthly advance	\$77,297	\$26,144
Monthly advance - Tenure	\$497	\$203
Creditline in 5 years	\$142,966	\$154,791
Creditline in 10 years	\$204,391	\$239,602

(1) Tied to 1-year CMT; (2) Lesser of FHA limit or home value; (3) 2% of maximum claim amount

A Borrower Could Get	Monthly Adjusting Loans	Annual Adjusting Loans
1) A single lump sum advance of	\$177,297	\$126,841
2) Or a creditline account of	\$177,297	\$126,841
that grows each year by	7.41%	9.13%
If unused, available credit		
in 5 years would be	\$253,744	\$196,339
in 10 years would be	\$362,380	\$303,915
3) Or a monthly loan advance for tenure	\$1,140	\$959
4) Or any combination of lump sum at closing, creditline account, and monthly advance		

Source: www.reversemortgage.org

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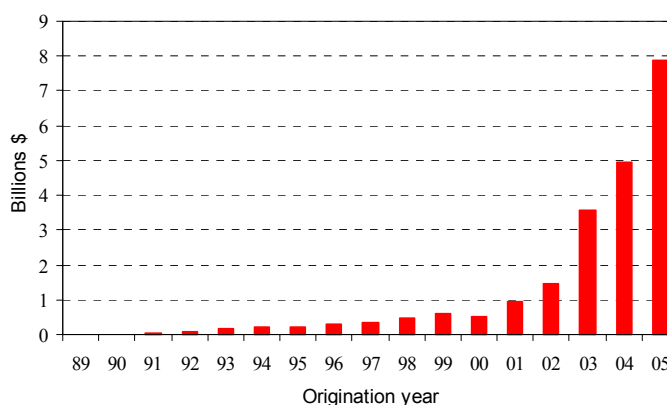
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III. HECM Borrower and Loan Characteristics

Since 2001, the total principal limit of HECM loans has grown more than eight-fold to a total volume of \$8 billion in 2005.

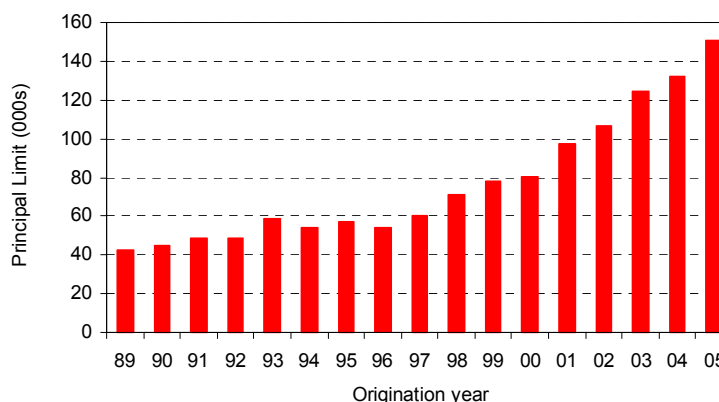
Based on a dataset of over 180,000 HECM loans originated since the program started in 1989, we now present some characteristics of HECM mortgages and borrowers. In Figure 7, we show the total principal limit at origination for HECM loans grouped by origination year. Since 2001, the program has experienced an eight-fold increase in volume. Thus, while the total principal limit of loans originated by HECM lenders was about \$1 billion in 2001, lenders originated loans with a total principal limit of almost \$8 billion in 2005. The rapid growth in the total principal limit of originated loans can be attributed to both the growth in the number of loans originated (Figure 2) and the increase in the average principal limit per loan (Figure 8). Figure 8 shows that the average principal limit of loans originated in 2005 was \$151,000 versus \$97,000 for 2001 originated loans, representing an increase of 60%. This increase is probably due to the very rapid growth in home price appreciation (HPA) over the last few years.

Figure 7: HECM Total Principal Limit by Origination Year



Source: HUD

Figure 8: HECM Average Principal Limit by Origination Year



Source: HUD

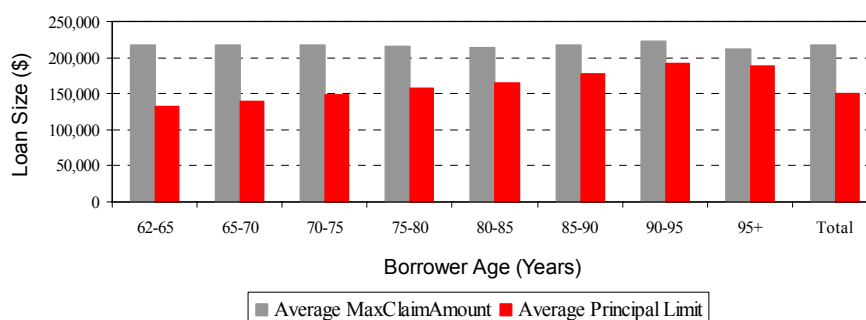
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The average principal limit is higher for more senior borrowers. However, the maximum claim amount is approximately the same for different age groups.

In Figure 9, we show the distribution of the average principal limit and the average maximum claim amount based on the borrower's age at origination for the HECM loans originated in 2005. The average maximum claim amount seems to be approximately constant among the different borrower age groups. Although one might expect more elderly borrowers to have greater equity in their homes and consequently a higher maximum claim amount, the FHA loan size limit probably restricts the variation in the maximum claim amount among the different age groups. However, as one would expect, the average principal limit is higher for more senior borrowers than relatively younger borrowers. Thus, more senior borrowers can receive a higher upfront payment or establish a larger line of credit than relatively younger borrowers.

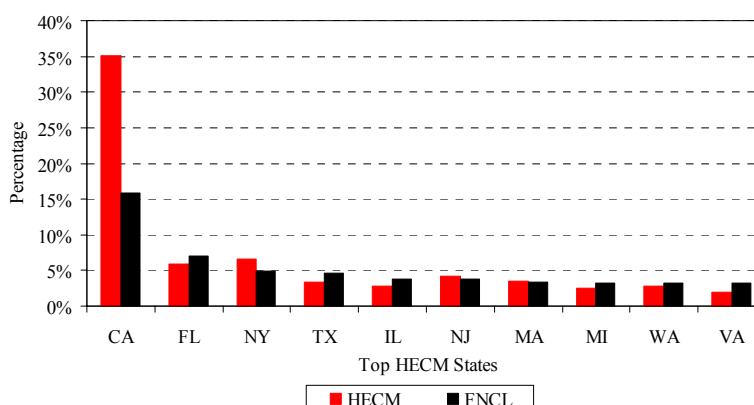
Figure 9: Principal Limit and Maximum Claim Amount Distribution for 2005 HECM Originations



Source: HUD

Approximately 35% of the HECM loans were originated in CA.

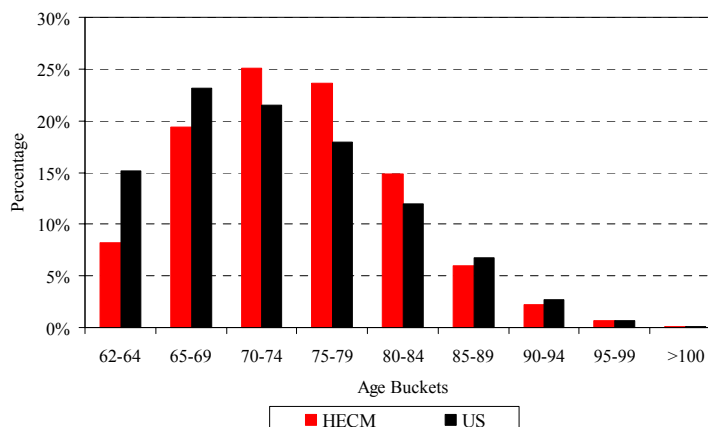
In order to understand the geographic distribution of HECM loans, we plot the percentage distribution (by principal limit) of HECM loans in the top 10 states with the most HECM loans and compare it with the distribution (by current balance) of FNMA 30-year loans in Figure 10. Approximately 69% of the originated HECM loans come from the 10 states shown in Figure 10 with 35% coming from the state of California alone. In comparison, only 16% of the FNMA 30-year loans are from California. The relatively high California concentration of HECM loans may be due to 1) a large number of retirees living in California; 2) higher home prices in California than in other U.S. states; and, 3) increased borrower awareness of HECM loans in California relative to other states.

Figure 10: The Geographic Distribution of HECM and FNMA 30-year loans

Source: Fannie Mae, HUD, Banc of America Securities

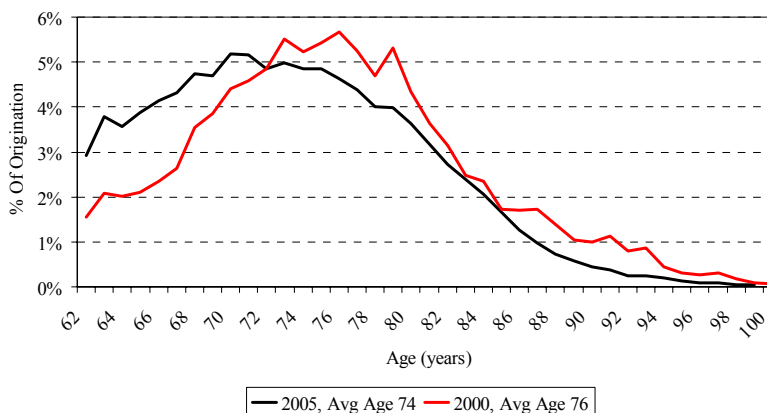
In 2005, the average age of an HECM borrower was 74 years.

In Figure 11, we graph the overall borrower age distribution for HECM loans versus the distribution for the U.S. population for ages greater than or equal to 62 years. The graph suggests that on average the population of HECM borrowers is older than the universe of eligible borrowers. For instance, while 21% of the US population over the age of 62 falls in the 70-74 age group bucket, 25% of the HECM borrowers are from this age group. Likewise, 24% of HECM borrowers are from the age group of 75-79, while the percentage of US population of age 62 or higher in this age group is only 18%. Figure 11 also suggests that HECM loans are relatively less popular among the 62-64 or 65-69 age group borrowers. Thus, while 15% of the borrowers who would potentially qualify for HECM loans fall in the 62-64 age group bucket, only 8% of the actual HECM borrowers are from this age group. However, it appears that the age distribution of HECM borrowers has shifted slightly towards the younger population group over the past few years (Figure 12). Consequently, while in 2000 the average age of a HECM borrower was 76 years, in 2005 the average age was close to 74 years.

Figure 11: Borrower Age Distributions for HECMs versus the U.S. Population (Age >= 62 years)

Source: U.S. Census Bureau (based on 2000 Census), HUD, Banc of America Securities

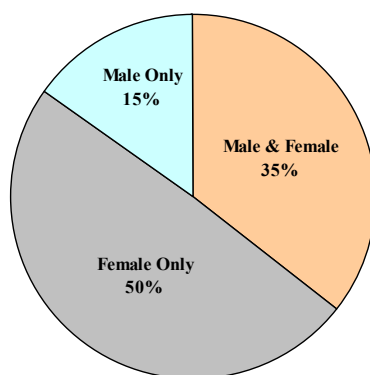
Figure 12: HECM Borrower Age Distribution by Origination Year



Source: HUD, Banc of America Securities

In addition to the age of the borrower, gender is also very important. Since females have lower mortality rates than males, a loan given to a female borrower is expected to stay outstanding for a longer time period than one given to a male borrower of the same age. In Figure 13, we show the distribution of HECM borrowers based on gender. 50% of the HECM borrowers fall under the “female only” category while the percentage of “male only” borrowers is only 15%. Approximately 35% of the HECM loans are taken jointly by male and female co-borrowers.

Figure 13: HECM Borrower Distribution by Gender



Female borrowers comprise a large share of HECM borrowers.

Source: HUD, Banc of America Securities

We now take a look at the repayment characteristics of the HECM loans. Based on the HECM data set, repayment reasons on HECM loans are documented based on the following four categories:

Mortality and mobility account for three quarters of loan repayment. Refinancing is not very common among HECM borrowers.

- Death of the borrower;
- Borrower moving out of the property;
- Voluntary repayment (i.e., the borrower paid off the loan but remained on the property); and
- “Other” causes for repayment.

Unfortunately, for a large fraction of the paid off loans the reason for loan repayment is not known. In spite of this data limitation, the known data lead us to several interesting conclusions. As one would expect, death is the primary cause for loan repayment (~51%), followed by “other” (~22%), mobility (~20%) and refinancing (~6%). Thus, the data suggests that mortality and mobility are the primary causes for repayment and that voluntary repayment is not very common among HECM borrowers.

IV. Repayment and Draw Behavior of HECM Loans

In this section, we present a simple framework that can be used to estimate repayments on HECM loans. In the absence of sufficient historical data, rating agencies often use a similar framework to rate securitized reverse mortgage deals. The approach presented here is based on estimating repayments on a pool of reverse mortgage loans by using borrower mortality and mobility rates. Following this, we present repayment seasoning curves for borrowers from different age groups as the repayment rate on reverse mortgages is observed to depend strongly on a borrower's age.

Finally, understanding how borrowers draw cash from their available credit-line is of considerable importance in estimating net repayments (i.e., repayments less the amount drawn) on a pool of reverse mortgage loans since a very large fraction of HECM borrowers select a LOC as their payment plan. We present a sample draw curve that illustrates borrower draw behavior for LOC HECM loans.

Repayment Behavior of HECM Loans

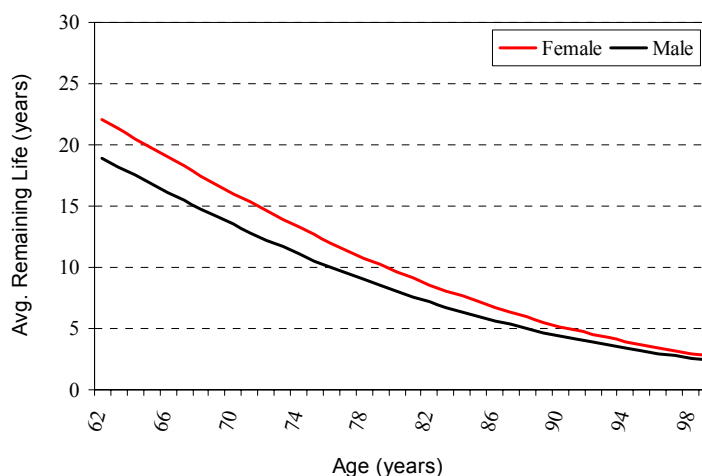
Some of the factors that lead to repayments on reverse mortgage loans are mortality, mobility, and refinancing. Based on our analysis of HUD data, mortality and mobility are probably the two most important factors. Refinancing is probably less important for repayment speeds because of high origination costs and because the interest rate on reverse mortgage loans is adjustable (monthly or annually). Therefore, a framework for estimating repayments based on just mortality and mobility rates should provide reasonable results.

The repayment rate on a pool of reverse mortgages is strongly determined by borrower mortality and mobility rates.

Mortality Rates

We begin by presenting data on life expectancy for the male and female population in Figure 14. The data shown here is from the 2003 United States Life Table produced by the U.S. Department of Health and Human Services. The data gives readers an estimate of the maximum time for which a reverse mortgage loan taken out by a male or female borrower of a certain age can be expected to stay outstanding. As expected, the data indicate that females have a higher life expectancy than males and that life expectancy decreases with age. Thus, in the absence of borrower mobility or voluntary refinancing, a reverse mortgage loan given to a 74 year old male borrower is expected to be repaid in 11 years on average due to the borrower's mortality. Similarly, the same loan given to a 90 year old male borrower under similar circumstances is expected to be repaid in 4.5 years on average. Note that in practice actual repayments will be faster than the rate obtained using borrower's mortality alone because of mobility (such as a borrower moving to a nursing home or an assisted living facility) and other reasons.

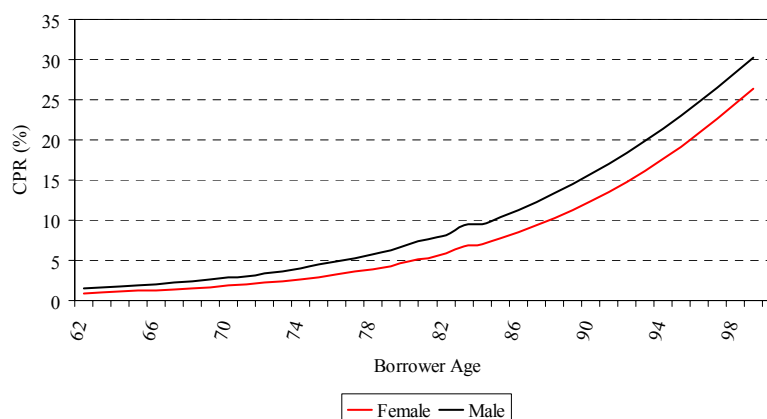
It is quite possible that due to self-selection, the average life expectancy of reverse mortgage borrowers is higher than that suggested by the data presented in the Life Table. Borrowers who know that they are in good health and who expect to live longer are more likely to pay the high origination costs associated with taking out a reverse mortgage. A similar self-selection bias is observed for life insurance policy holders and insurance companies have developed their own life expectancy tables to account for such biases.

Figure 14: Male and Female Life Expectancy

The repayment rate due to mortality increases with age.

Source: United States Life Tables, 2003

The annualized mortality rate for male and female U.S. population is shown in Figure 15. The data obtained from the 2003 United States Life Table represents the probability of death within the next 12 months for a person of a given age. If we define the repayment speed (CPR, %) due to mortality as the number of loans repaid in a year divided by the number of loans outstanding at beginning of the year, then the annualized mortality rate presented in Figure 15 is equivalent to the repayment speed due to borrower mortality. The data indicate that repayment rate due to mortality increases with age and that females have a lower mortality related repayment rate than males.

Figure 15: Annualized Mortality Rate of the U.S. Population

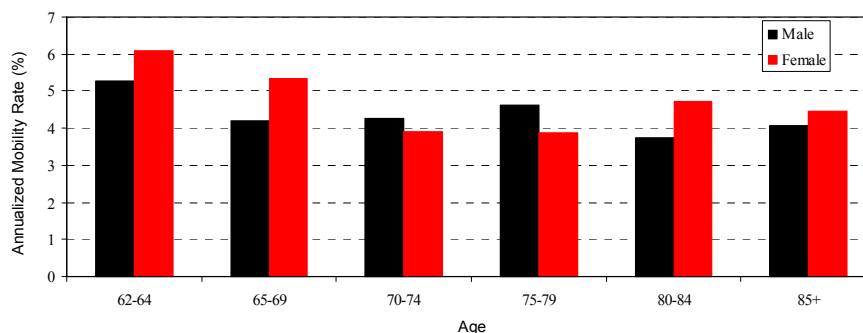
Source: United States Life Tables, 2003.

Mobility Rates

Figure 16 shows U.S. Census Bureau data for the annualized mobility rate of Americans grouped by age and gender. It is interesting to observe that although female borrowers in the age group of 62-69 have low mortality rates, they have relatively high mobility rates. In general, mobility is first observed to decrease with an increase in age and then increase for the more senior population group. The increase in mobility for the higher age group is perhaps

because of health related concerns that may force these seniors to move out of their homes to nursing homes or assisted-care facilities.

Figure 16: Annualized Mobility Rate of American Population



Source: U.S. Census Bureau, Current Population Survey, 2004 Annual Social and Economic Supplement

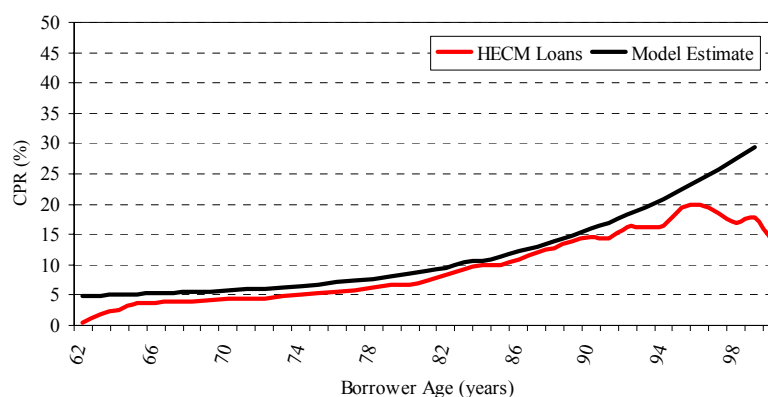
Combining Mortality and Mobility Rates

By combining the mortality and mobility data presented in Figures 15 and 16, one can obtain a rough estimate of repayment speeds on reverse mortgages for borrowers of different ages. For a given age, the probability that a borrower will not die over the following 12 months is equal to $(1 - \text{Annualized Mortality Rate})$. Similarly, the probability of a borrower not moving is equal to $(1 - \text{Annualized Mobility Rate})$. The probability of the loan not repaying over the next 12 months can then be obtained as $(1 - \text{Annualized Mortality Rate}) * (1 - \text{Annualized Mobility Rate})$. Repayment speeds on reverse mortgages can then be calculated as follows:

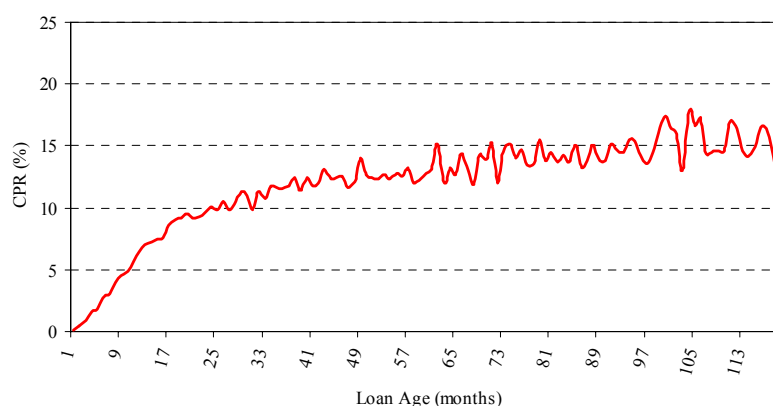
$$\text{CPR} = 1 - [(1 - \text{Annualized Mortality Rate}) * (1 - \text{Annualized Mobility Rate})]$$

To illustrate this approach, we compute the repayment rates implied by the mortality and mobility data using the mortality rate for female borrowers and assuming a constant mobility rate of 4% for simplicity. In Figure 17, we compare the computed repayment speeds with the actual repayment experience on our HUD HECM data set. It is quite remarkable that results from this simplified model do a reasonably good job of predicting the general trends in actual repayment speeds. Clearly, the model consistently overestimates actual repayment speeds. There are probably several factors behind the difference but the overall issue is that we are looking at two very different sample populations. The U.S. Census Bureau numbers are calibrated to a sample of the entire U.S. population, whereas the HECM data reflect a much narrower borrower base.

In particular, the data presented in Figure 17 has not been controlled for loan age. Figure 18 shows an aggregate seasoning curve for HECM loans based on actual data. The repayment speed calculation here is the fraction of loans that repay in a year versus the loans outstanding at the beginning of the year. As is typical with prepayments for regular mortgages, there is a seasoning ramp associated with reverse mortgages. However, reverse mortgage loans seem to season over a longer time period than do the regular mortgages. Based on the historical data, overall HECM speeds are observed to stabilize at around 15% CPR after 5 years.

Figure 17: HECM Aggregate Repayment Speeds by Borrower Age

Source: HUD, Banc of America Securities

Figure 18: HECM Aggregate Repayment Speeds by Loan Age

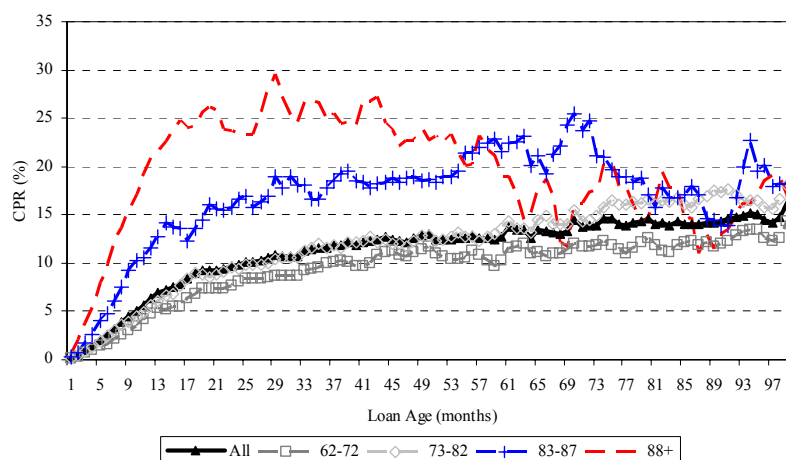
Source: HUD, Banc of America Securities

HECM loans seem to season over a longer time period than regular mortgages. Aggregate HECM speeds appear to stabilize around 15% CPR after 5 years.

To illustrate the effect of borrower age on repayment speeds, we show the historical seasoning curve for HECM loans grouped by borrower age at origination in Figure 19 along with aggregate repayment speeds. Borrowers in the age group of 62-72 (i.e., younger borrowers) have speeds slower than aggregate repayment speeds, while the (more senior) borrowers in the age groups of 83-87 and 88+ have repayment speeds that are faster than the aggregate. Another interesting observation that we can draw from Figure 19 is that the more senior borrowers have a steeper seasoning ramp (24-36 months) than the seasoning ramps for relatively younger borrowers. The drop in speeds for the 83-87 age group borrowers after about 72 months and for the 88+ age group borrowers after about 48 months is probably because of the lack of sufficient historical data. It is also possible that there could be a drop in repayment speeds in these age groups due to “burnout”: Borrowers in this age group who have not repaid in 48-72 months are the borrowers who are in relatively good health and are expected to live longer.

Figure 19: HECM Repayment Speeds by Loan Age & Borrower Age at Origination

Older HECM borrowers have a steeper seasoning ramp.

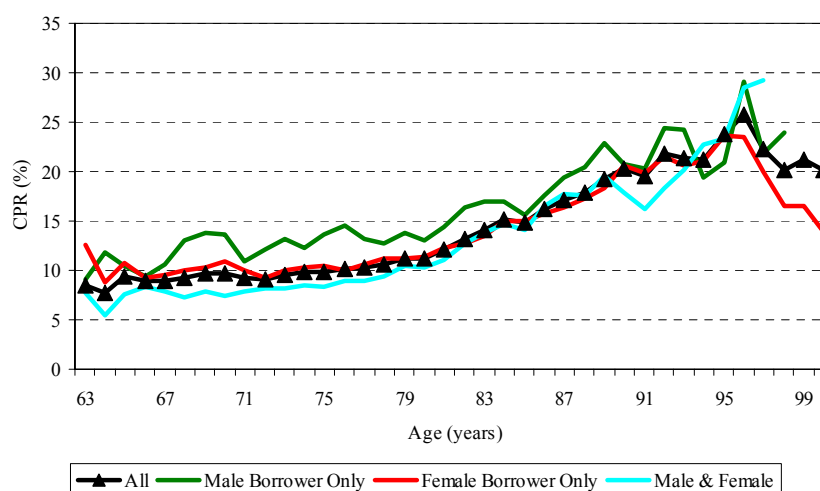


Source: HUD, Banc of America Securities

Figure 20 shows a plot of fully seasoned HECM repayment speeds grouped by the current age and gender of the borrower. The figure indicates that male borrowers repaid the fastest while the male and female borrower category had the slowest rate of repayment. Note that for male and female co-borrowers, repayment due to mortality is after the death of both borrowers. The combined probability of survival of the co-borrowers is higher than their individual survival probabilities. Therefore, we should expect that repayments due to mortality for male and female co-borrowers should be lower than mortality related repayments on male only and female only borrowers. Furthermore, several studies have shown that marriage has a beneficial effect in prolonging the lives of the married couple.

Figure 20: Fully Seasoned Repayment Speeds by Borrower Age for HECM loans

Male borrowers have the fastest fully seasoned repayments, while the “male and female” borrower category has the slowest rate of repayment.



Source: HUD, Banc of America Securities

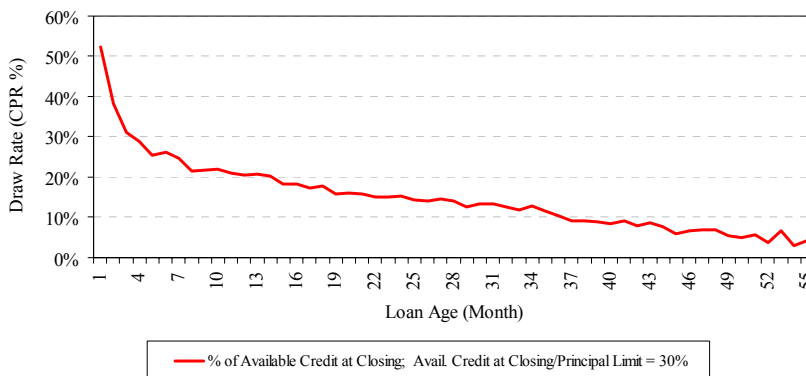
Draw Rate for Line of Credit (LOC) HECM Loans

The prepayment analysis presented so far provides useful information to investors and lenders in estimating the number of loans out of a large pool that would prepay over any given time period. For LOC HECM loans, it is also important to understand how borrowers withdraw funds from their LOC as well. As we will see in the next section on structuring HECM loans, the issuer usually provides a reserve fund at the inception of a structured deal that is used to fund a HECM borrower's withdrawal from their established LOC. These reserve funds are usually invested in interest bearing securities. The rate of withdrawal from the reserve fund and the interest earned on it can have important implications on principal repayment for securitized tranche holders.

The draw rate on a collection of LOC HECM loans can be defined in different ways. One way is to define the draw rate as the annualized dollar amount withdrawn per month as a percentage of the available credit at closing. The second way to define it is as the annualized dollar amount withdrawn per month as a percentage of the original loan balance at closing. The advantage of the former approach is that the available credit at closing is a good indicator of the maximum amount that an issuer has to put in a reserve account to fund future withdrawals by borrowers. Based on a set of HECM loans from a reverse mortgage originator, approximately 37% of LOC borrowers take a lump sum distribution equal to their entire principal limit at origination. The average closing line of credit balance as a fraction of the borrower's principal limit at origination is 27%.

Figure 21 shows a hypothetical draw curve as a percentage of the available credit at closing for LOC HECM loans. Note that the available credit at closing is some fraction of the principal limit since borrowers typically take out most of their principal limit (~70%) at loan closing. The remaining portion of their principal limit is the available line of credit that gets withdrawn as shown in Figure 21. The figure shows that in the initial few months after loan origination, the rate of withdrawal from the available line of credit is between 25% to 50% and then slowly decreases over time to less than 10% after the first 36 months and then to 2%-3% after 50 months.

Figure 21: Generic Draw Curve as % of Available Credit at Closing



Source: Banc of America Securities

V. Basic Concepts in Structuring HECMs

It's not straightforward to see how to drive a securitization based on reverse mortgage payments. Some of the key issues associated with structuring line of credit (LOC) HECM loans can be summarized as follows:

- During the first few years, the amount withdrawn by borrowers from their line of credit can be greater than repayments on the underlying mortgage pool. Any securitization of reverse mortgage loans needs to provide a mechanism to fund this excess of draws over repayments on the underlying loans in these initial years. After this initial period, the repayment rate on the underlying loans will be sufficient to fund borrower draws on their credit lines.
- Even if there are no draws in a given month, a mortgage insurance premium that equals 0.5% of the outstanding loan balance has to be paid to HUD for insuring the loans. Furthermore, a servicing fee is paid monthly to servicers. The securitization mechanism should provide enough cash to pay for monthly mortgage insurance premiums and servicing fees even when there are no net repayments on the underlying mortgage loans.
- The trust issuing securities supported by reverse mortgage loans needs to pay current interest to AAA rated security holders even if there were no net positive repayments on the underlying mortgage loans in a given month.²

To satisfy these obligations, a trust issuing securities backed by reverse mortgage loans sets up a *funding account* and deposits cash into it. To illustrate the amount of cash that the funding account needs to hold and the amount of securities that a trust can issue, we walk through a simple example. Let's assume that we have a pool of 1,000 LOC reverse mortgage loans with an aggregate principal limit of \$100mm. Assume that the total outstanding balance of the LOC reverse mortgage loans is, on average, 70% of the principal limit at inception, the available credit on closing is 27% of the principal limit, and the remaining 3% is the servicing fees set aside. This gives us a total outstanding balance on the loan pool of \$70mm (average outstanding balance of \$70,000 per loan), a net available line of credit of \$27mm, and a \$3mm servicing fee set aside. Note that the borrowers can withdraw their entire line of credit at any time. Therefore, at inception the funding account should hold a minimum of \$27mm in cash plus other amounts required to be set aside by the rating agencies. In this scenario, the trust can issue \$100mm of securities, acquire \$70mm reverse mortgage loans and put the remaining \$30mm in a funding account.

Depending upon the repayments and draws on the reverse mortgages, a funding account that holds \$30mm at inception should be sufficient to satisfy all obligations that an issuing trust has to different counterparties. For example, it is fairly typical for draws on the loans to exceed repayments for the first twelve months. Beginning in about the thirteenth month, repayments on mortgage loans start to exceed net draws. Therefore, the funding account will

² AAA-rated security holders would prefer to get paid interest in a timely fashion but the securitization structure may not guarantee timely payment of interest.

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need to pay the net negative draw by borrowers in the first year but in most cases will not be needed for this purpose beyond that.

In addition to being able to satisfy the demands of HECM borrowers, the funding account should also have sufficient cash to pay interest to security holders. Let's assume that security holders receive interest at the rate of 1-month Libor + 20 bps on their \$100mm of principal. Also, assume that the \$30mm in the funding account can be invested in 1-month Libor. At a constant 1-month Libor rate of 5.30%, this implies that the payments to the security holders in a year (\$5.5mm) exceed the interest earned on the funding account (\$1.6mm) by \$3.9mm. This excess interest will have to be paid from the funding account and will further deplete it.

Finally, even when net repayments on the mortgage loans are positive (i.e., starting month 13), they may not be sufficient to pay interest owed to security holders over the next few years and this would further reduce the balance in the funding account. However, when the reverse mortgage loans collateralizing the trust are sufficiently seasoned, net repayments should be sufficient to satisfy both interest and principal payments on the securities.

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